Risk Management Plan (RMP):

1. Safety specification

1.1 Data and approach

I described the 320 reports by product and reaction, then classified each reaction. A listed (known) reaction was called an important identified risk when at least 15% of its reports were serious and at least one was a hospitalisation. This is my working rule, and a medical reviewer should confirm it. A reaction that is not listed but has serious reports was called a potential risk. Severity (mild, moderate, severe) is not stated in the pack and is kept separate from seriousness.

1.2 Reports by product and reaction

Product	Reaction	Reference status	Reports	Serious (%)	Hosp.	Other med. imp.	Not recovered	Age 65+
Amlodipine	Peripheral oedema	Known	51	10 (19.6%)	5	5	11	16
Amoxicillin	Rash	Known	39	1 (2.6%)	0	1	9	12
Atorvastatin	Myalgia	Known	46	7 (15.2%)	3	4	15	20
Ciprofloxacin	Tendon pain	Known	35	7 (20.0%)	3	4	9	11
Losartan	Fatigue	Unexpected	38	3 (7.9%)	1	2	12	8
Metformin	Nausea	Known	33	3 (9.1%)	1	2	10	8
Omeprazole	Headache	Known	42	8 (19.0%)	4	4	4	11
Sertraline	Dizziness	Known	36	9 (25.0%)	3	6	9	11
Total			320	48 (15.0%)	20	28	79	97

Hosp. = hospitalisation. Other med. imp. = other medically important condition. Outcome is unknown in 80 reports (25%), so the not recovered column is a minimum. 97 reports (30%) are in patients aged 65 or over, and 18 of the 48 serious reports (38%). Without exposure by age, this cannot be read as a higher risk in older patients.

1.3 Important identified risks

Risk	Evidence in the pack	Comment
Ciprofloxacin: tendon pain	35 reports, 7 (20.0%) serious, 3 hospitalisation. Listed reaction. 11 reports aged 65 or over.	Tendon damage is the concern behind this reaction, but the pack records tendon pain only. Rupture is not recorded and cannot be assessed.
Sertraline: dizziness	36 reports, 9 (25.0%) serious (highest share), 3 hospitalisation. Listed reaction. 13 reports with a dose of 500 mg.	Falls or injury are not recorded. Older patients (11 reports aged 65 or over) need particular attention in the materials.
Amlodipine: peripheral oedema	51 reports, 10 (19.6%) serious, 5 hospitalisation (the most of any product). Listed reaction.	13 reports have a dose above the usual maximum (500 mg or 20 mg). Dose-related effects cannot be judged until doses are verified.
Omeprazole: headache	42 reports, 8 (19.0%) serious, 4 hospitalisation. Listed reaction.	9 reports have a dose of 500 mg. Headache is non-specific, so other causes are possible.
Atorvastatin: myalgia	46 reports, 7 (15.2%) serious, 3 hospitalisation. Listed reaction. 20 reports aged 65 or over (43%, the largest share).	Pack records myalgia only. Muscle damage such as rhabdomyolysis is not recorded and cannot be assessed.

Other identified risks. Amoxicillin rash (39 reports, 1 serious, no hospitalisation) and metformin nausea (33 reports, 3 serious, 1 hospitalisation) are listed but fall below the threshold. Routine pharmacovigilance is enough for them.

1.4 Important potential risks

Losartan: fatigue. Fatigue is not listed in the Synthetic Core Data Sheet v1.0. There are 38 reports, 3 serious (1 hospitalisation, 2 other medically important). The number of reports does not show that losartan causes fatigue. Fatigue is common and non-specific, and underlying illness or other medicines could explain it. The signal needs a case-by-case evaluation (activity A1).

Medication errors and use outside the usual indication. 52 reports (16%) give a single dose above the usual adult daily maximum for the product, such as 500 mg of amlodipine, atorvastatin, losartan, omeprazole or sertraline. 104 reports (33%) give an intravenous route for products that, as far as I know, are oral only. 246 reports (77%) give an indication that does not match the usual use of the product, and 276 (86%) have at least one of these. Because so many reports are affected, much of this may be data capture error and not real prescribing. It must be verified before it is treated as a true risk (activity A3).

Product	Reports	Dose above usual maximum	IV route (oral-only products)	Indication mismatch	At least one of these
Amlodipine	51	13	22	41	44
Amoxicillin	39	0	n/a	29	29

Product	Reports	Dose above usual maximum	IV route (oral-only products)	Indication mismatch	At least one of these
Atorvastatin	46	11	21	39	43
Ciprofloxacin	35	0	n/a	27	27
Losartan	38	6	22	22	32
Metformin	33	0	18	23	30
Omeprazole	42	9	n/a	37	38
Sertraline	36	13	21	28	33

Usual doses and routes are from standard references and should be confirmed against the product information. The pack gives the dose without a frequency, so a single dose is compared with the daily maximum.

1.5 Missing information

Item	Evidence in the pack	Proposed action
Children and adolescents	Youngest patient is 18 years. No reports under 18.	Follow-up in the post-authorisation safety study (A2)
Pregnancy and breastfeeding	No field in the reports	Enhanced follow-up of exposed pregnancies (A4)
Renal and hepatic impairment	No field in the reports	Targeted questionnaire for serious reports (A4)
Long-term use	Longest time to onset is 35 days. No treatment duration is given.	Cohort follow-up beyond 35 days (A2)
Concomitant medicines and interactions	Not captured in the aggregate file	Targeted questionnaire and review in A2

2. Pharmacovigilance plan

2.1 Routine pharmacovigilance activities

Activity	What it covers
Case collection and processing	All sources, including consumers, pharmacists and physicians. MedDRA coding, seriousness and expectedness against the Synthetic Core Data Sheet v1.0, and causality assessment.
Expedited reporting	Serious cases within 15 calendar days (7 days for fatal or life-threatening unexpected cases). Non-serious cases are reported in periodic reports or within the local deadline. Check current local rules.
Signal detection	Continuous review of the safety database, with a monthly review of listed and unlisted reactions.
Periodic safety update reports	Every 6 months for the first 2 years after launch, yearly for the next 2 years, then every 3 years, unless the EU reference date list sets a different frequency.
Literature monitoring	Screening of the medical literature at least weekly.
Targeted follow-up questionnaires	For serious reports of the five important identified risks and for losartan fatigue. They collect dates, dose, duration, medical history, concomitant medicines, dechallenge, rechallenge and outcome.
Case data quality checks	Validation rules at case entry for dose, route and indication, and review of the 276 reports already flagged.

2.2 Additional pharmacovigilance activities

ID	Activity	Concern	Objective and approach	Proposed milestones
A1	Signal evaluation	Losartan: fatigue	Review all 38 reports case by case, search the literature and medical review of the 3 serious cases. Decide whether to add fatigue to the reference data sheet.	Start when the RMP is approved. Report with the first periodic safety update.
A2	Non-interventional post-authorisation safety study (cohort)	Five important identified risks; older patients; long-term use; children	Follow treated patients in a real-world database to measure how often the risks occur with exposure figures, including by age and treatment duration.	Protocol 3 months after launch. Yearly progress reports. Final report about 3 years after launch.
A3	Data verification, then drug utilisation study	Medication errors and use outside the indication	First verify the 276 flagged reports with reporters. If confirmed, describe actual dose, route and indication in practice. The study also measures the effect of the risk minimisation tools.	Verification within 3 months. Protocol at 6 months. First results at 12 months.
A4	Enhanced follow-up in special populations	Pregnancy, breastfeeding, renal and hepatic impairment	Targeted questionnaires for every report in these groups, with a cumulative review.	Start at launch. Review in every periodic safety update.

A2 and A3 are studies that the RMP requires (category 3 under EU rules). All milestones are proposals and would be agreed with the regulator.

3. Risk minimisation measures

3.1 Routine measures

Product information (summary of product characteristics, sections 4.4 and 4.8, and the package leaflet) listing the known reactions and how to act on them, clear labelling of dose and route, and prescription-only supply. The legal status is not stated in the pack, so prescription-only is my assumption.

3.2 Additional measures

Tool	Audience	Key messages	Distribution
Patient alert card (ciprofloxacin)	Patients taking ciprofloxacin	Stop the medicine and contact a doctor straight away if you have tendon pain or swelling. Rest the area. Do not change the dose. Show the card to any doctor or pharmacist.	Handed out with each supply by the dispensing pharmacy. Full text in Appendix A.
Educational brochure for prescribers and pharmacists	Prescribers and pharmacists for all eight products	Check the indication, dose and route before prescribing or dispensing. Take extra care in older patients. Ask about tendon pain, muscle pain, dizziness, swelling and headache. Fatigue with losartan is being evaluated. How to report. Outline in Appendix B.	Sent by the marketing authorisation holder before launch and again at launch, then on request.
Patient brochure	Patients and carers	Take the medicine exactly as prescribed and do not change dose or route. Symptoms to report and when to seek help. How to report a side effect.	Given with the first dispensing.

The materials must be agreed with the regulator before launch.

3.3 Measures by safety concern

Safety concern	Routine measures	Additional measures
Ciprofloxacin: tendon pain	Product information	Patient alert card, prescriber and pharmacist brochure, patient brochure
Sertraline: dizziness	Product information	Prescriber and pharmacist brochure (older patients), patient brochure
Amlodipine: peripheral oedema	Product information	Prescriber and pharmacist brochure (dose), patient brochure
Omeprazole: headache	Product information	Prescriber and pharmacist brochure (dose)
Atorvastatin: myalgia	Product information	Prescriber and pharmacist brochure, patient brochure (report muscle pain)
Losartan: fatigue	Product information	None until the signal evaluation (A1) is complete
Medication errors and use outside the indication	Clear labelling of dose and route	Prescriber and pharmacist brochure; drug utilisation study (A3)
Missing information	Product information	None. Addressed by A2 and A4

4. Effectiveness measures

Measure	Type	Indicator	Baseline from the pack	Proposed target
Distribution	Process	Share of target prescribers and pharmacists who received the brochure	Not available	At least 90%
Knowledge	Process	Share who answer the key messages correctly in a survey	Not available	At least 80%
Alert card use	Process	Share of ciprofloxacin patients who say they received the card	Not available	At least 80%
Dose	Outcome	Share of reports with a dose above the usual maximum	52 of 320 (16%), unverified	Lower at 12 months than the verified baseline
Route	Outcome	Share of reports with an intravenous route for oral-only products	104 of 320 (33%), unverified	Lower at 12 months than the verified baseline
Serious tendon pain	Outcome	Serious reports per unit of exposure, from study A2	7 of 35 ciprofloxacin reports serious, no exposure figures	No increase over the study period

The baselines are not reliable until the flagged reports are verified. Targets are proposals to be agreed with the regulator.

5. Assumptions and limitations

- Product: the pack names no single product, so the eight products are treated as one portfolio. Marketing authorisation holder, launch date and indications are unknown.
- No exposure figures are available, so no incidence or reporting rate was calculated. Report counts were not used to infer causation, as the README requires.
- Seriousness is taken as reported. The reason for each serious classification is not stated. Severity is not stated.

- The 15% serious threshold for important identified risks is my working rule and needs medical review.
- Usual doses and routes come from standard references and should be confirmed. The pack gives no dose frequency.
- Hosp. = hospitalisation. Other med. imp. = other medically important condition. Outcome is unknown in 80 reports. Dechallenge and rechallenge are inconsistent in places. No exact duplicate reports were found (0), but true duplicates cannot be excluded because the file has no patient identifiers.
- There is no clinical trial, non-clinical or epidemiology data in the pack, so those modules are not completed and benefit-risk is not assessed.
- Expectedness is judged only against the synthetic reference data sheet. All cases are fictional, and this overview is not for regulatory submission.